svm derivation and margin

x = [1,5,6]

equations calculate w, b

w_1*x[1]_1 + w_2*x[1]_2 + b = 1 w_1*x[5]_1 + w_2*x[5]_2 + b = -1 w_1*x[6]_1 + w_2*x[6]_2 + b = -1

w_1*1 + w_2*1 + b = 1 w_1*1 + w_2*0 + b = -1 w_1*0 + w_2*1 + b = -1

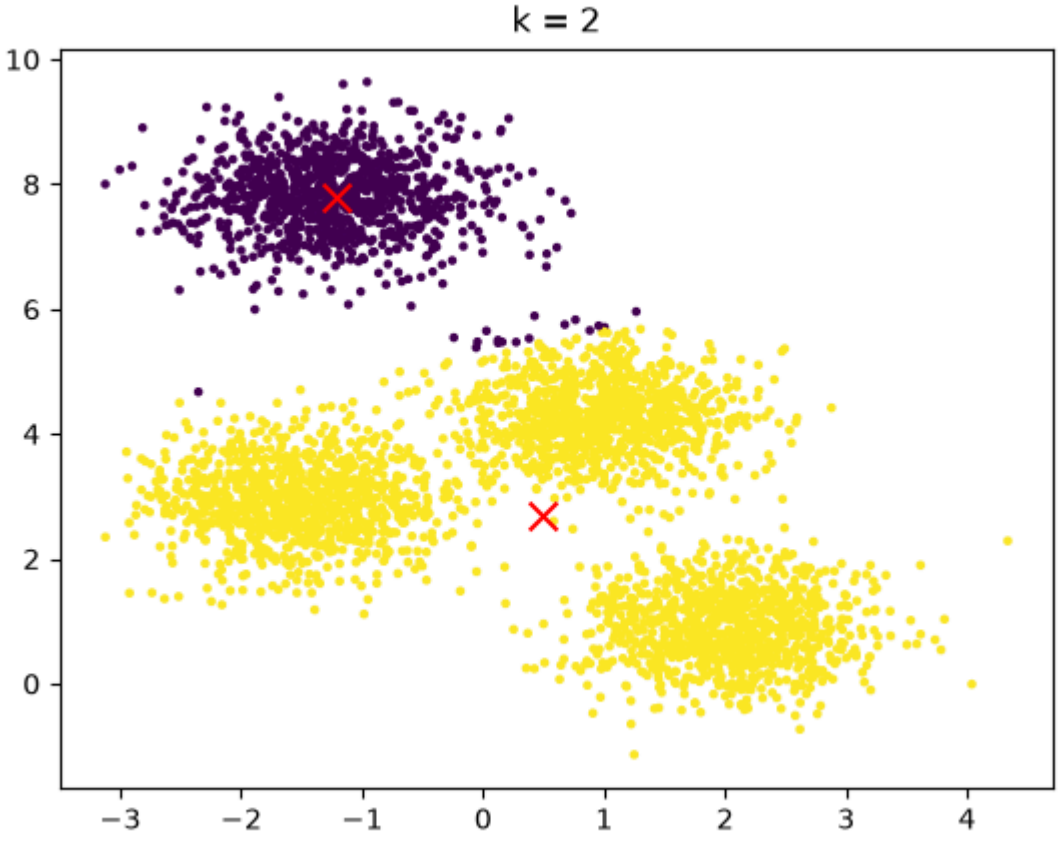
w_1 + w_2 + b = 1 w_1 + b = -1 w_2 + b = -1

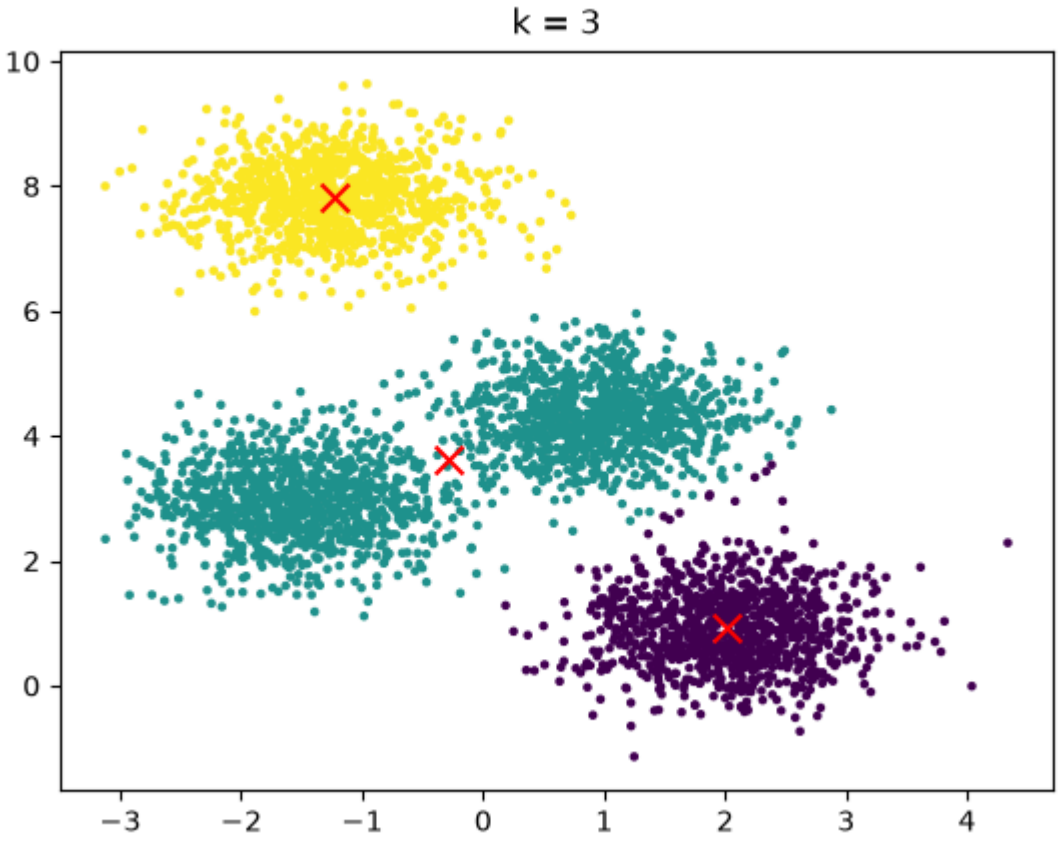
b = -w_1-w_2+1 w_1 = -1-b w_2 = -1-b

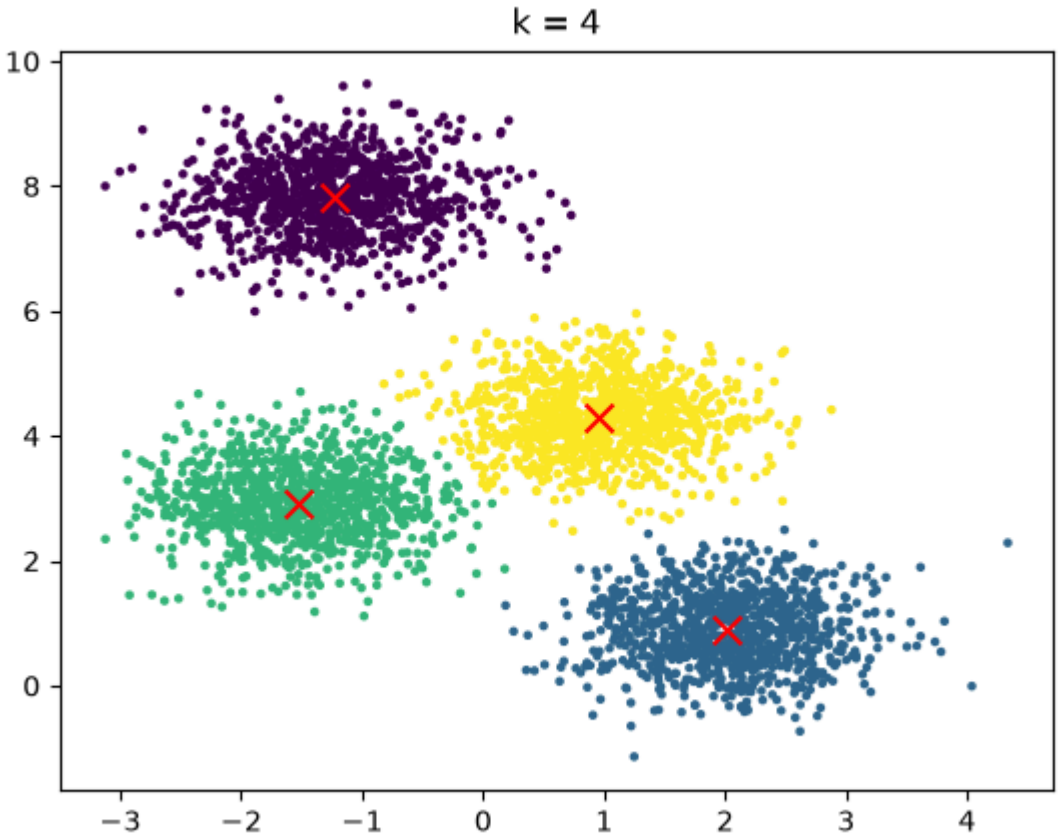
b = -(-1-b)-(-1-b)+1 b = -(-2-2b)+1 b = 2+2b+1 b = -3

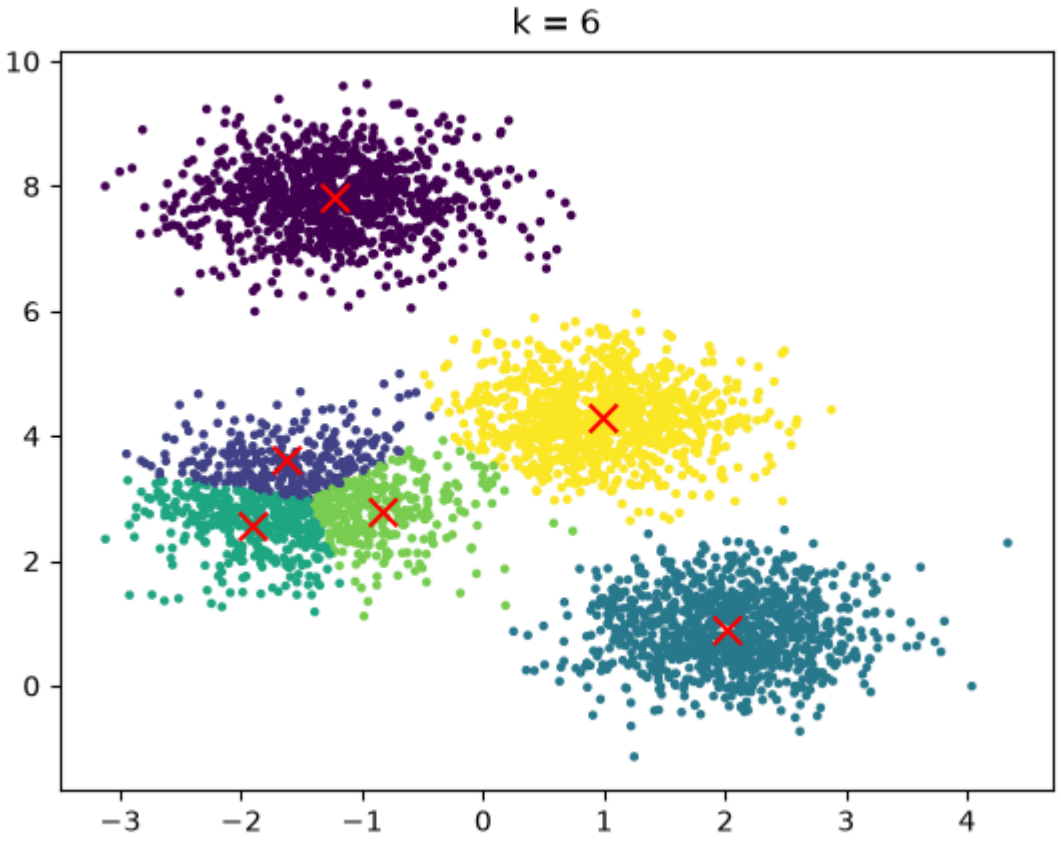
b = -3 w_1 = -2 w_2 = -2

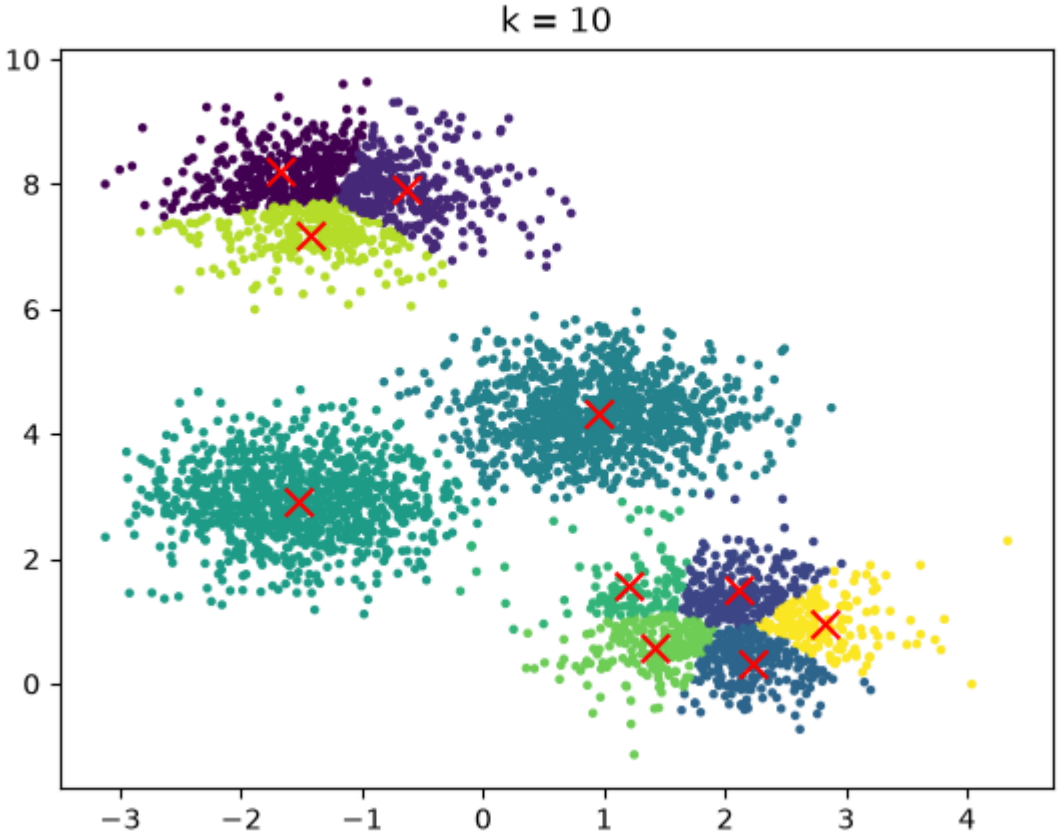
k-means plots and discussion


k = 2
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


k = 3
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


k = 4
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4


k = 6
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4



As k increases, the cluster means start to define the 4 groups, after 4 clusters, the clusters start getting overfit, splitting a group into multiple different clusters

two method accuracy comparison

```
Mean baseline accuracy:      0.8279500000000001
Mean K-NN repaired accuracy: 0.8644000000000001
Change from baseline:        0.03644999999999998
```

experiment csv

csv file